

Clinical Policy: A Critical Issue in the Management of Adult Patients Presenting to the Emergency Department With Acute Carbon Monoxide Poisoning (Executive Summary)

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From the American College of Emergency Physicians Clinical Policies Subcommittee (Writing Committee) on Carbon Monoxide Poisoning:

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BACKGROUND

This clinical policy from the American College of Emergency Physicians addresses key issues in the evaluation and management of adult patients presenting to the emergency department with acute carbon monoxide poisoning. A writing subcommittee conducted a systematic review of the literature to derive evidence-based recommendations to answer the below critical question. For this question, a systematic literature search was performed, evidence was graded and synthesized, and recommendations were made based on the strength of the available data.

CRITICAL QUESTION

In ED patients diagnosed with acute CO poisoning, does HBO2 therapy, compared with normobaric oxygen therapy, improve long-term neurocognitive outcomes?

Patient Management Recommendations

Level A recommendations. None specified.

Level B recommendations. None specified.

Level C recommendations. In symptomatic CO poisoning, selected patients may benefit from HBO2 treatment based on severity of symptoms and availability (distance and time).

Translation of Classes of Evidence to Recommendation Levels

Based on the strength of evidence for the critical question, the subcommittee drafted the recommendations and supporting text synthesizing the evidence using the following guidelines:

Level A recommendations. Generally accepted principles for patient care that reflect a high degree of scientific certainty (eg, based on evidence from one or more Class of Evidence I or multiple Class of Evidence II studies that demonstrate consistent effects or estimates).

Level B recommendations. Recommendations for patient care that may identify a particular strategy or range of strategies that reflect moderate scientific certainty (eg, based on evidence from one or more Class of Evidence II studies or multiple Class of Evidence III studies that demonstrate consistent effects or estimates).

Level C recommendations. Recommendations for patient care that are based on evidence from Class of Evidence III studies or, in the absence of adequate published literature, based on expert consensus. In instances where consensus recommendations are made, “consensus” is placed in parentheses at the end of the recommendation.

Relevant industry relationships: *There were no relevant industry relationships disclosed by the subcommittee members for this topic.*

Relevant industry relationships are those relationships with companies associated with products or services that significantly influence the specific aspect of disease addressed in the critical question.